

Eric Ming Tang, Ph.D.

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TECHNICAL SKILLS (Experience)

Programming Languages (8 years): Python, C/C++ (multithreaded, real-time), MATLAB, Java

Machine Learning & AI (6 years): Deep learning (PyTorch/LibTorch, TensorFlow), CNNs (U-Net, YOLOv4), GANs; real-time object detection, classification, segmentation, and tracking; dataset curation and annotation QC; model training, evaluation, and optimization

Edge Deployment: PyTorch model export via ONNX, TensorRT optimization for latency-constrained real-time inference on embedded hardware (NVIDIA Jetson Orin Nano); integration into low-latency C++ runtimes (LibTorch)

Computer Vision (8 years): Geometric computer vision (camera calibration, stereo vision, 3D reconstruction; OpenCV); signal and image processing, denoising, super-resolution, multimodal image fusion

Camera & Imaging Systems (8 years): Multimodal camera and sensor integration (visible-light, near-infrared OCT, photoacoustic); 3D/4D imaging systems; multi-view acquisition; camera sensor characterization; devices from prototype through validation

Software Tools (8 years): Microsoft Visual Studio, VS Code, Qt Creator, LabView, OpenCV, Scikit-learn

Hardware (8 years): Control systems, DAQ/FPGA programming, hardware control APIs (cameras, scanners, lasers, sensors), optical system design and optimization

EDUCATION

Vanderbilt University, Ph.D. in Biomedical Engineering 2022

Duke University, Bachelor of Science in Biomedical Engineering 2018

Duke University, Bachelor of Arts in Computer Science 2018

WORK EXPERIENCE

Topcon Healthcare San Jose, CA
Senior Research Scientist May 2023 – Present

- Managed and optimized real-time image acquisition software (C++) for high-throughput data capture at 2 gigasamples/second and flexible customization of imaging workflows via multithreaded programming
- Designed a physics-informed deep learning model (PyTorch) for image super-resolution and denoising, integrating the optical system's PSF and acquisition parameters, achieving a 2x improvement in resolution and signal quality; results were presented to company executives and stakeholders
- Deployed deep learning imaging models to embedded edge hardware by exporting PyTorch models via ONNX and optimizing with TensorRT for real-time inference on an NVIDIA Jetson Orin Nano
- Led cross-organizational collaborations between vendors and engineering teams to co-develop a novel compact imaging sensor and scale it from development to production, achieving a 2x reduction in device size; contributed to hardware integration, software evaluation, and quantitative benchmarking
- Evaluated and enhanced third-party analysis software by identifying algorithmic limitations, technical problem solving, providing performance feedback, and iterating improvements

Diagnostic Imaging and Image-Guided Interventions Laboratory Nashville, TN
Postdoctoral Research Fellow May 2022 – May 2023

- Extended and adapted a deep-learning-based pipeline for 4D surgical instrument tracking by implementing multi-channel inputs and data augmentation, improving model robustness and accuracy in diverse imaging conditions and achieving 99% sensitivity at 23 fps
- Investigated advanced machine learning models (MATLAB/Python: multiscale context aggregation networks, pix2pix, cGANs) to improve neural network input quality by up to 4x under challenging surgical conditions
- Supported development, training, and real-time deployment of an image denoising framework utilizing multi-scale U-Net architectures to improve image quality by 90% with video-rate performance at 22 fps

Graduate Research Assistant May 2018 – May 2022

- Optimized a high-speed multimodal imaging system and managed acquisition software codebase (C++) that supported real-time processing, live preview of reflectance and 3D images, and GPU-accelerated CNN architectures at 4 gigasamples/second
- Integrated an automated deep-learning-based framework for surgical instrument tracking using multi-view inputs via a lightweight model (YOLOv4) for real-time image acquisition and object detection at 120 fps

SELECTED PUBLICATIONS ([Google Scholar](#))

1. **Tang, E. M.**, El-Haddad, M. T., Patel, S. N., and Tao, Y. K., "Automated instrument-tracking for 4D video-rate imaging of ophthalmic surgical maneuvers," *Biomedical Optics Express* 13(3), 1471–1484 (2022).
2. Rico-Jimenez, J. J., Hu, D., **Tang, E. M.**, Oguz, I., and Tao, Y. K., "Real-Time OCT Image Denoising Using Self-Fusion Neural Network," *Biomedical Optics Express* 13(3), 1398–1409 (2022).
3. **Tang, E. M.** and Tao, Y. K., "Modeling and optimization of galvanometric point-scanning temporal dynamics," *Biomedical Optics Express* 12(11), 6701–6716 (2021).

PROJECTS

ericmtang.com ("TangOS") — a personal website with the look and feel of a desktop OS: draggable windows, dock with magnification, and a document-style resume viewer. Built from scratch with Next.js, React, and TypeScript. Source: github.com/tangericm/tang-os